National University of Sciences and Technology

School of Electrical Engineering and Computer Science



Final Year Project Guidelines

Foreword

This is the Final Year Project handbook of the School of Electrical Engineering and Computer Science. This handbook contains guidelines for the conception, preparation, implementation, completion and finally the assessment of Final Year Projects.

The intention of this handbook is to develop guidelines and a uniform structure and outline for undergraduate students. It serves as an instructional manual for the expected contents, deliverables, quality and the required quantity of the final projects for students and also provides evaluation rubrics for supervisors and evaluators.

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Revision History

Compiled By	Date	Reason For Changes	Version
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Dr. Omar Arif	10th Oct, 2016	Updated poster rubrics	1.1
Ms. Maryam Sajjad	20th Feb, 2017	Updated the assigned numbers on criteria of each graded item	1.2
Dr. Anis Ur Rahman	6th Mar, 2017	Updated marks distribution	1.3
Dr. Muhammad Muneeb Ullah	2nd Nov, 2017	Updated FYP timeline	1.4

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1.1 Introduction

The Final Year Project (FYP) is the culmination of students' degree program. The main purpose of this project is to encourage students to apply the knowledge acquired during their studies. It allows them to work on a substantial problem for an extended period of time, show how proficient they are in solving real world problems. It brings them a sound opportunity to demonstrate their competence as professionals and to apply what they have learnt in the other components of the degree. Besides, they get a chance to improve their technical skills, communication skills by integrating writing, presentation and learn how to work in teams. With a real-world problem at hand, they get to learn professional practice and a variety of non-technical issues such as management, finance, safety, reliability, environment and social impacts. Moreover, it provides an integrated assessment of the progress of the students toward the training they went through during their academic tenure at the college.

FYP course is different other courses because it demands independent objective formulation, planning, management and self-motivation. It is therefore essential to design fair and comprehensive guidelines for the students, supervisors and the evaluators. A structured manual and lifecycle process is therefore essential in order to help students conform to the required quality standards, outline general expectations from the supervisors and sketch assessment criteria for the evaluators. Hence, contribute as a fundamental underpinning to achieve high quality learning outcomes of the projects.

1.2 Degree Program Learning Outcomes (PLOs)

Program Learning Outcomes are the narrower statements that describe what students are expected to know and be able to do by the time of graduation. These relate to the knowledge, skills and attitude that the students acquire while progressing through the program. The program must demonstrate that by the time of graduation the students have attained a certain set of knowledge, skills and behavioral traits, at least to some acceptable minimum level. Specifically, it is to be demonstrated that the students have acquired the following graduate attributes:

(i) **Engineering Knowledge:** An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

(ii) **Problem Analysis:** An ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

(iii) **Design/Development of Solutions:** An ability to design solutions for complex engineering problems and design systems, components or processes

that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

(iv) **Investigation:** An ability to investigate complex engineering problems in a methodical way including literature survey, design and conduct of experiments, analysis and interpretation of experimental data, and synthesis of information to derive valid conclusions.

(v) **Modern Tool Usage:** An ability to create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering activities, with an understanding of the limitations.

(vi) **The Engineer and Society:** An ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solution to complex engineering problems.

(vii) **Environment and Sustainability:** An ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.

(viii) **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

(ix) **Individual and Team Work:** An ability to work effectively, as an individual or in a team, on multifaceted and /or multidisciplinary settings.

(x) **Communication:** An ability to communicate effectively, orally as well as in writing, on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

(xi) **Project Management:** An ability to demonstrate management skills and apply engineering principles to one's own work, as a member and/or leader in a team, to manage projects in a multidisciplinary environment.

(xii) **Lifelong Learning:** An ability to recognize importance of, and pursue lifelong learning in the broader context of innovation and technological developments.

1.3 Overview of Final Year Project

A Final Year Project is a two-semester course in which students usually of 2-3 members select a project and are supervised by a faculty member. In this course, students choose a project subject and define the objectives of the project under the supervision of a faculty member, and prepare the project proposal including: defining the statement of the problem, defining system requirements, defining different candidate solutions for the problem of study, making feasibility study for different candidate solutions, defining the best candidate solution, defining time table schedule. Students present the final project report at the end of the semester to an evaluation a committee.

1.4 FYP milestones and evaluation stages

The FYP milestones, deliverables, evaluation stages along with their timelines is given the Table 1 and Table 2.

Table 1: Milestones & Marks Distribution

Proposal Defense (PD)	**5%**
○ Project Proposal Document + Presentation	(Advisor 60% of PD, Co-Advisor 40% of PD)
Design Expo (DE)	**10%**
○ DE Committee Marks	○ 10%
Mid Defense (MD)	**30%**
○ Advisor + Co-Advisor Marks (SRS)	○ 15% (8% Advisor+ 7% Co-Advisor)
○ KG's Marks (Presentation)	○ 15%
Final Defense(FD)	**45%**
○ KG's Marks for Presentation + Demonstration	○ 20%
○ Advisor's Marks (Presentation + Demonstration)	○ 10%
○ Advisor + Co-Advisor Marks (Report)	○ 10% + 5%
Open House	**10%**
Total	**100%**

Table 2: Detailed timeline of FYP phases

Phase	Week	Duration	Semester
Project & Committee Selection	Week 09	3 Weeks	6th
Proposal Defense	Week 15	1 Week	6th
Design Expo	Week 15	1 Week	7th
Mid Defense	Week 02	1 Day	8th
Final Defense	Week 14	1 Week	8th
Final Report Submission	Week 14		8th
Open House	Week 15	1 Day	8th

1.4.1 Proposal Defense

The students must choose a project subject and define the objectives of the project under the supervision of a faculty member, and prepare the project proposal. The format of FYP proposal defense and evaluation forms and their rubric are given in Annex A. Also the template for FYP proposal is added in Appendix A.

1.4.2 Mid-Defense/ Design Expo

After proposal defense, next FYP activity is Mid-Defense and Design expo. Students have to present their work and progress. The format of mid defense and evaluation forms and their rubric are given in Annex B. Students also have to submit the Software Requirement Specifications (SRS) document at the time of mid defense. Template for SRS and evaluation rubrics are attached in Annex B.

1.4.3 Final Defense

Final Defense is the final activity for FYP students, which is evaluated by Advisor, Co-advisor and evaluation committee. The format of FYP Final defense and evaluation forms and their rubric are given in Annex C. Also the template for FYP final report submission is attached in Annex C.

1.4.4 Open House

Open house is arranged for final exhibition of final year projects. Most of the people from industry visits FYP stalls and judges are allocated to each FYP for evaluation.

1.5 Guidelines for Project Supervision

Following rules should be taken under considerations during project supervision.

a) Each group will work under the supervision of an assigned supervisor throughout the final year (term 7 & 8).

b) Students are recommended to meet with their supervisor at least once a week. The students are expected to discuss their progress with their supervisors in these weekly meetings. Depending on students' requirements and the supervisor's availability, supervisors may also arrange additional meetings (physical/online) as requested.

c) Supervisors might also arrange communication with student groups via email or other means for the purpose of advising project groups.

d) It is the responsibility of the supervisor to inform his students with this handbook and all the included instructions and regulations.

1.5.1 Tasks expected from supervisors

During these meetings supervisors are expected to:

✓ To provide FYP Outlines / Objectives	Orientation
✓ Discuss project expectations and the plan with the group	
✓ To share previous practice experience, research, skills and expertise	
✓ Assign /Recommend related literature	Provide Knowledge
✓ Give training sessions on the respective research area and tell them what they need to know	
✓ To clarify students queries effectively as needed	
✓ To make students aware of professional ethics and standards	
✓ To advise students on how to deal effectively as a team while working under pressure, remaining optimistic and persistent, and how to meet milestone deadlines	
✓ To monitor the project progress on a weekly/fortnightly basis	Assess
✓ To ensure students are completing outlined project deliverables	
✓ To grade students work (at individual/group level) at the end of each semester	

1.5.2 Project Development Life Cycle:

The supervisors will guide the group through different steps in the software engineering life cycle and describe, discuss, assign, receive and review the corresponding outcomes/artifacts at the end of each step as described in Figure 1.

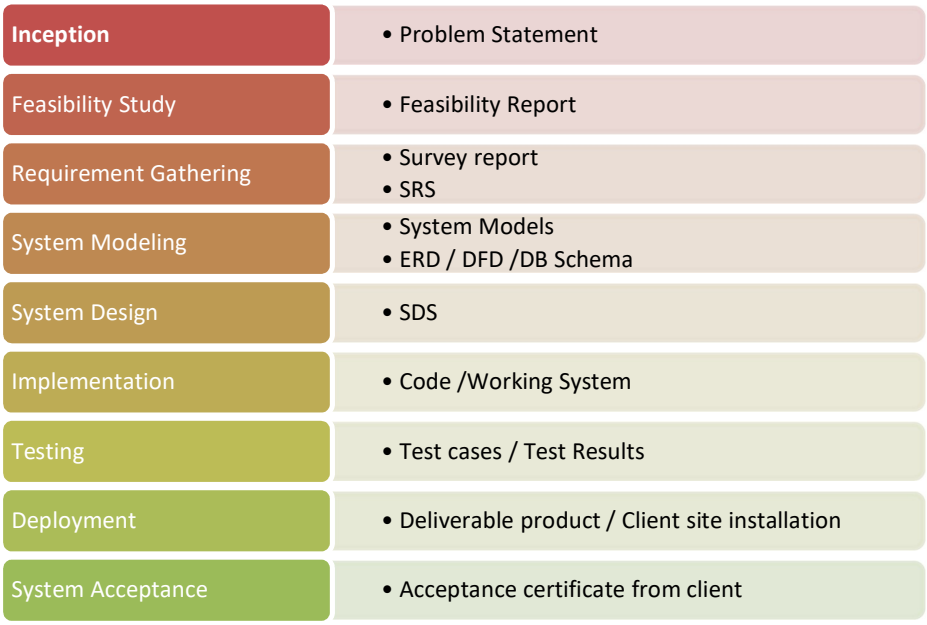


Figure 1: Software Engineering Lifecycle and Respective artifacts

During the Project Proposal, students undertake the initial phases of project planning, selection, analysis and designing phases. In the Project Implementation, students proceed with the implementation phase of their proposed project. As part of SDLC, the supervisors should guide the students to follow, but not limited to, the following best-practices:

a) Having a life cycle or system development methodology
b) Ensure proper research and background knowledge is acquainted
c) Feasibility study is conducted on the proposed project
d) Scope of the project is precise and crystal clear
e) Generating and comparing alternative designs to determine best match for the requirements
f) Roles & responsibilities of individual student working within the group is clear and accepted
g) Able to apply project resources as per the approved project plan
h) Track and report any issues and risks in completing assigned tasks
i) Both logical and physical design aspects are analyzed
j) Proper programming standards are maintained during the development of the project
k) Auto or Manual Test Cases are implemented and executed
l) Source control with versioning tools are used for developing as a team
m) Documenting required deliverables using industry standards
n) Participating in Seminars, Events, Publications and Workshops relevant to the project

1.6 Team Leadership:

Every graduation project group is assigned with a team leader who is essentially a cross-functional key player working within the project group. It is extremely important to get the right student within each group fit for the role of team leader which is crucial for the success of any project. Team leader should work very closely with the supervisor with the following, but not limited to, the primary tasks:

a) Provide input on the performance of team members
b) Resolve any conflicts within group members and maintain healthy group dynamics
c) Inform supervisor of any task delays and meeting hours change requests from students
d) Ensure rest of the group understands their roles and responsibilities on the assigned tasks;
e) Coordinate with internal or external project stakeholders on behalf of the team
f) Provide weekly status report – completed and signed by each team member as per the schedule

Note: During the course of the project, if the supervisor finds team leader is not performing as per the above expectations, the supervisors can request for the replacement of team leader with an alternative group member fit for this role.

1.7 Students Responsibility:

During the Final Year Project, students are responsible for the following:

a.	Agree with their assigned supervisors on the topic
b.	Perform weekly tasks, assigned by the supervisor (or distributed by the team leader)
c.	Discuss problems and seek advice from the supervisor in order to accomplish the assign tasks.
d.	Provide supervisor weekly status reports and get his/her feedback
e.	Apply recommendations to refine the previous task
f.	Finalize the project proposal and implementation, incorporating all the feedbacks and comments provided by the supervisor and evaluators.
g.	Conduct presentation at the end of each semester and defend project to the evaluation panel

1.8 Late submissions:

It is the responsibility of each group to ensure they complete the milestones of each semester and submit deliverables by the cutoff submission date. No project will be accepted after the cutoff date and necessary actions will be taken as per the supervisor and the evaluation committee decision policy.

1.9 Plagiarism:

Each project must be the original work of student groups. At the end of each semester, students will be required to present their project proposal and implementation outcomes as per the provided deliverables guidelines and the original work undertaken throughout each semester.

In the project report, for instance, if students have taken ideas or referencing other work as part of the proposed project, then, it must be cited and reference should be clearly specified. Same is the case while implementing the proposed solution. For instance, if students are developing project using 3rd party tools and libraries, it must be referenced and relevant comments and notes must be highlighted and will not be regarded as part of the original work of student groups. Hence, it is extremely important to note that it is the responsibility of students to ensure they are not plagiarizing knowingly or unknowingly.

In order to prevent plagiarism related issues, students are encouraged to get familiar with plagiarism specified in [1] and general referencing guidelines specified in [2].

In order to prevent plagiarism related issues during implementation, students are strongly encouraged to get familiar with software plagiarism specified in [3].

If students are found plagiarizing either in project proposal report or in the project implementation solution/code, immediate strict action will be taken as per the university policy.